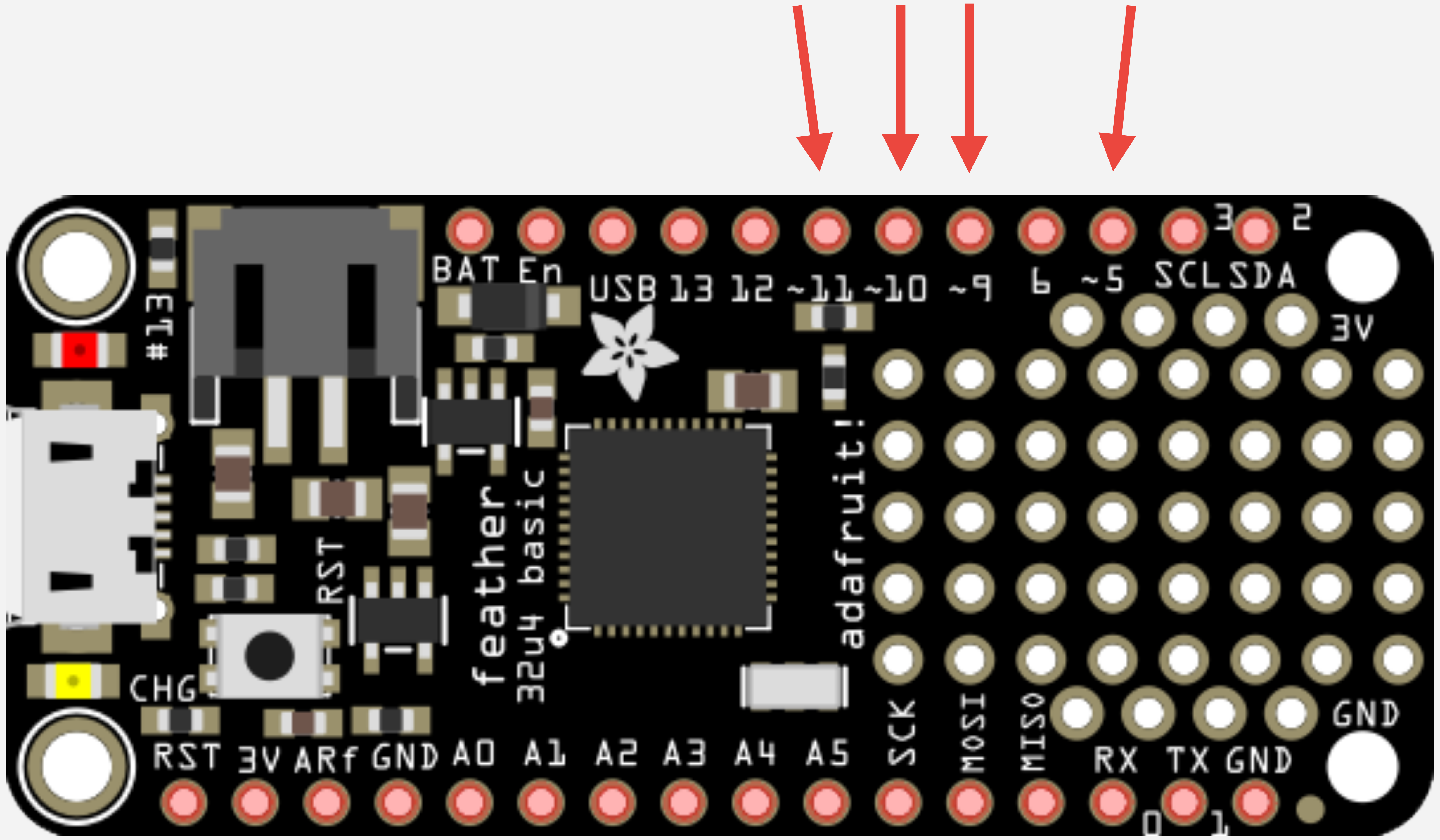
ANALOG OUTPUT AND BASIC AUDIO

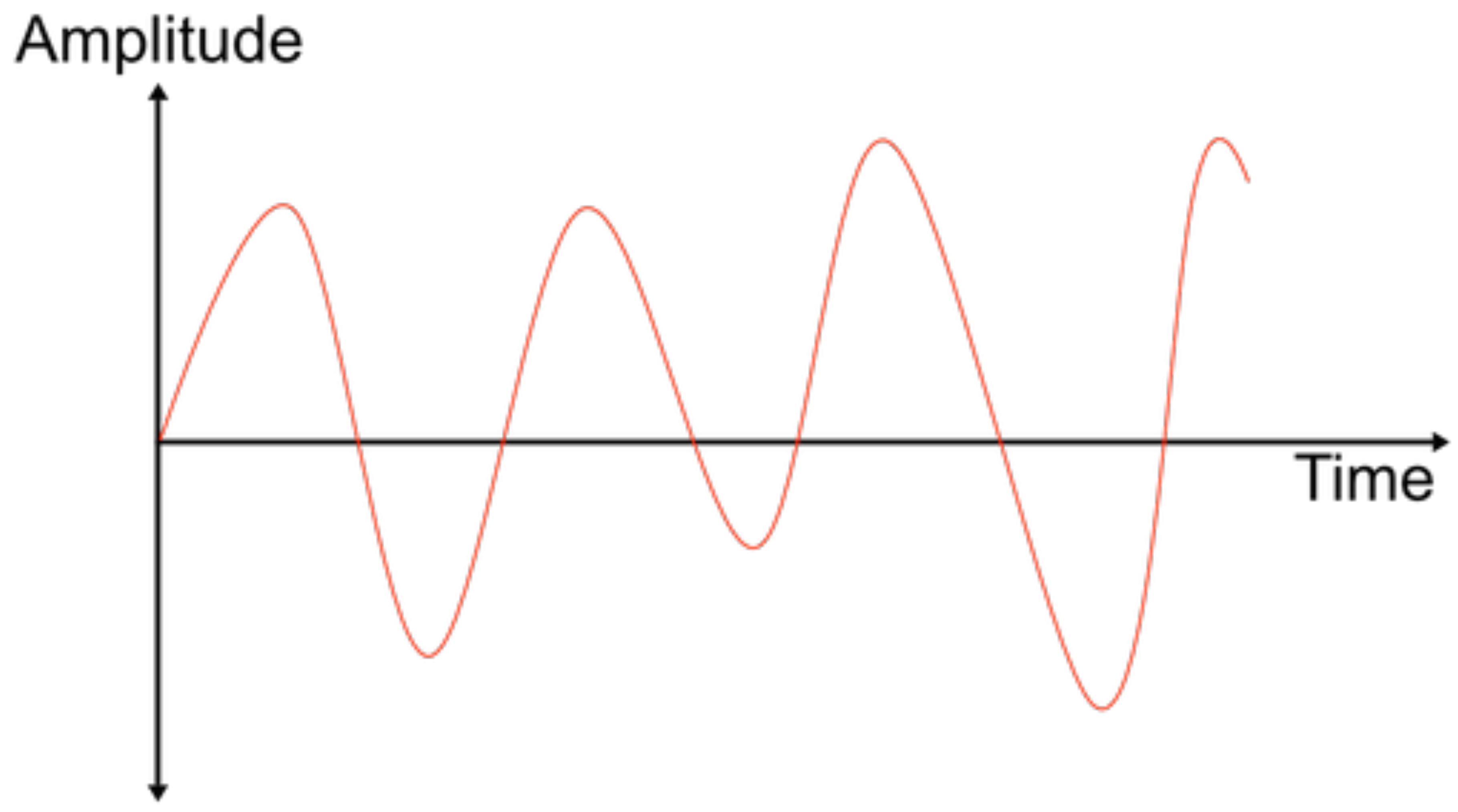
ANALOG OUTPUT

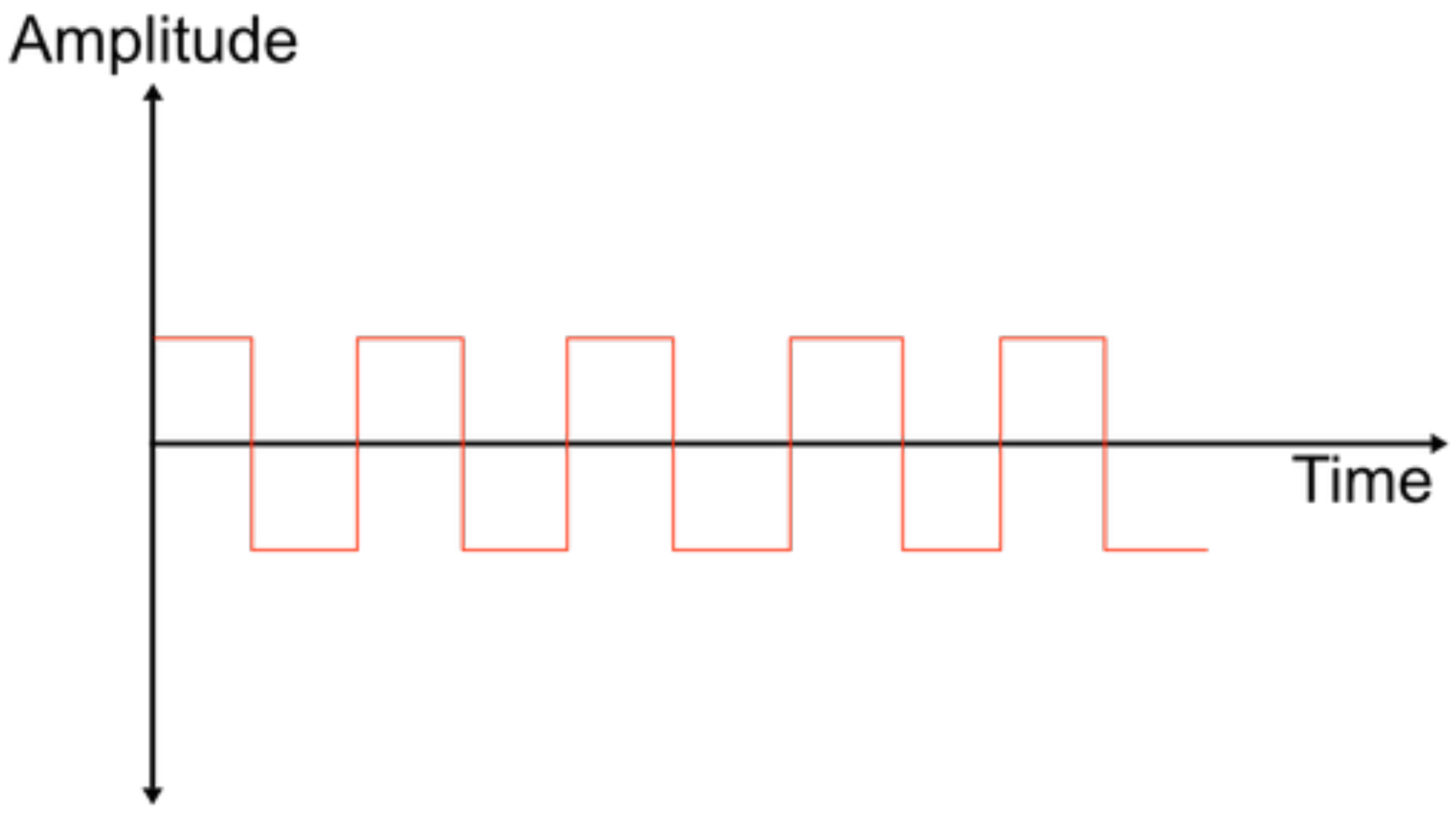


Analog Output*

*Only on the pins with the ~ next to them

Analog vs Digital



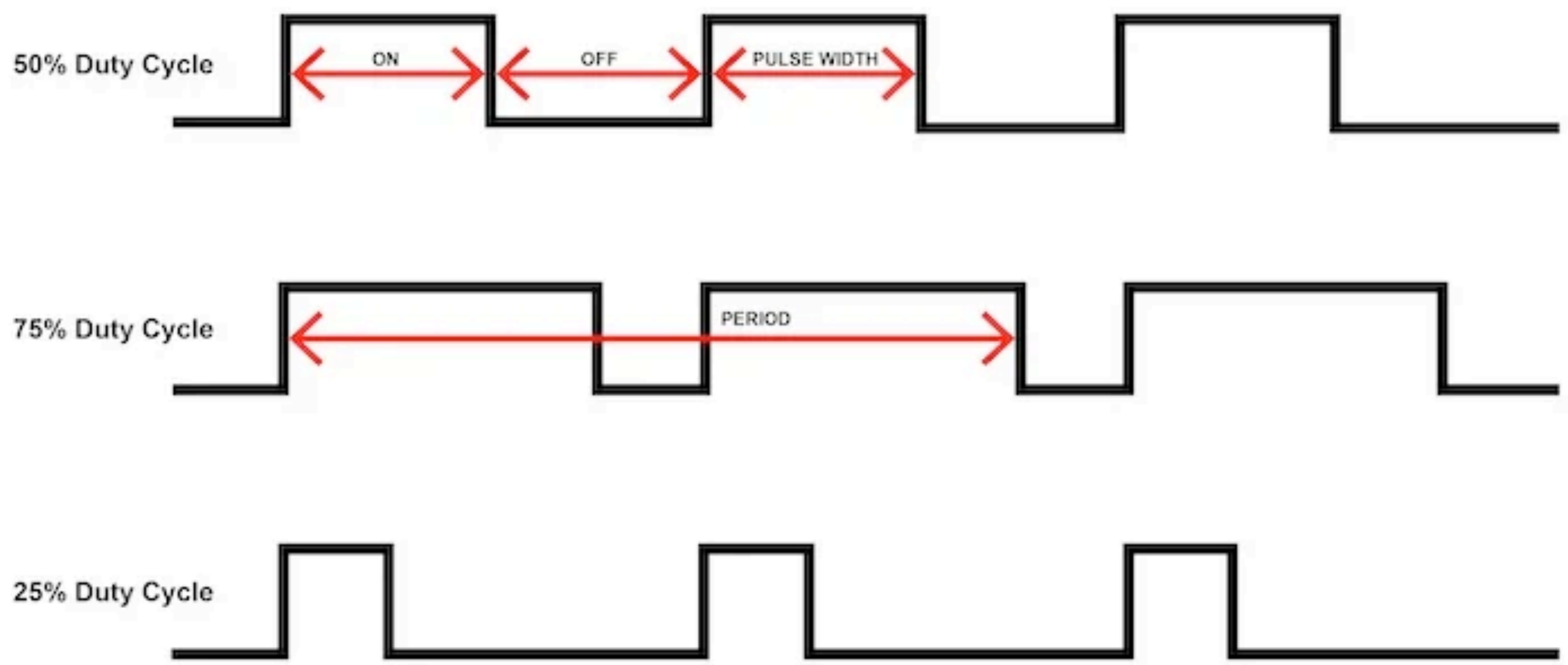


Analog Signal
Smooth gradient of infinite values.

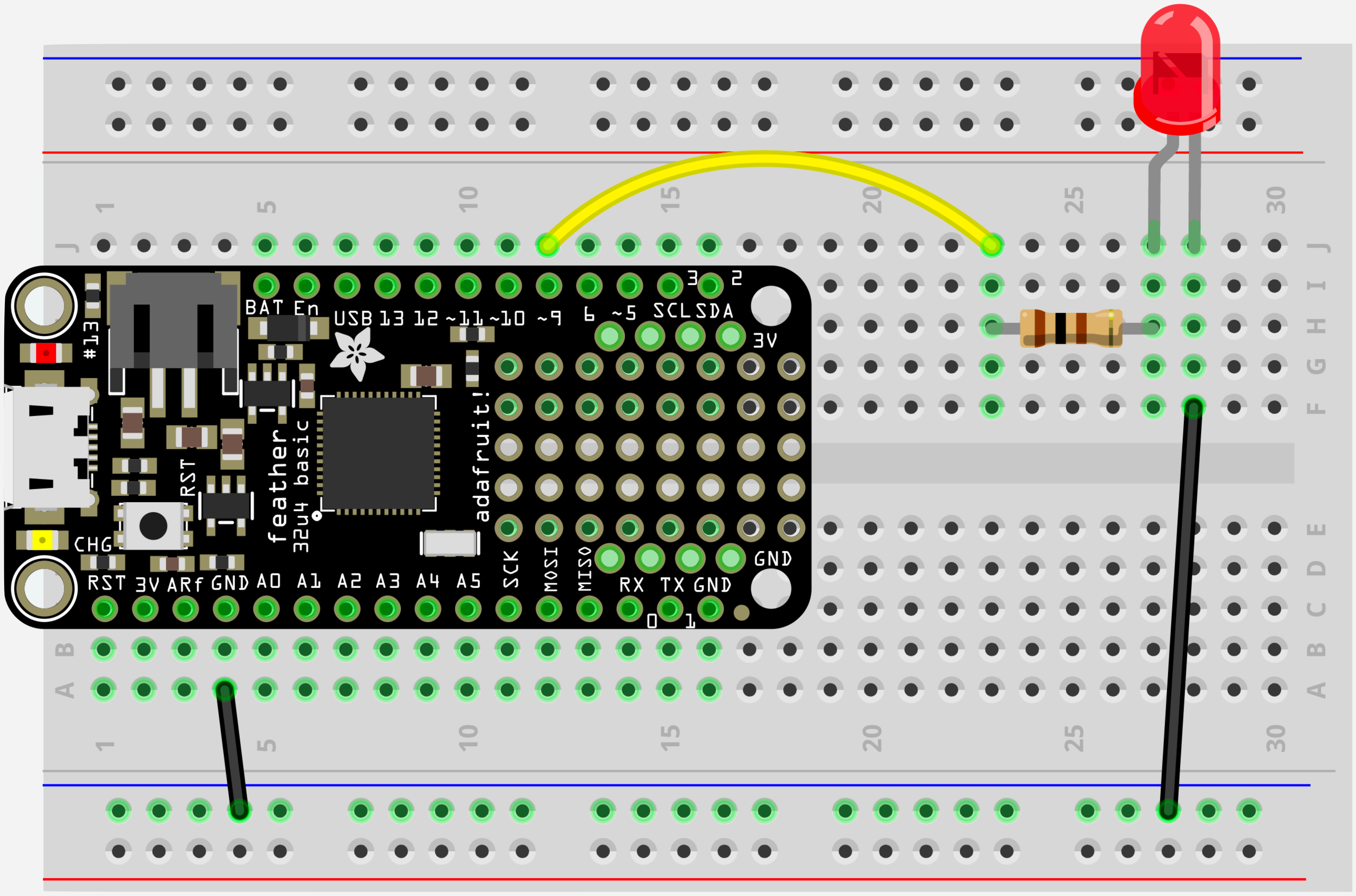
Digital Signal
Only 2 possible values.

PWM – Pulse Width Modulation

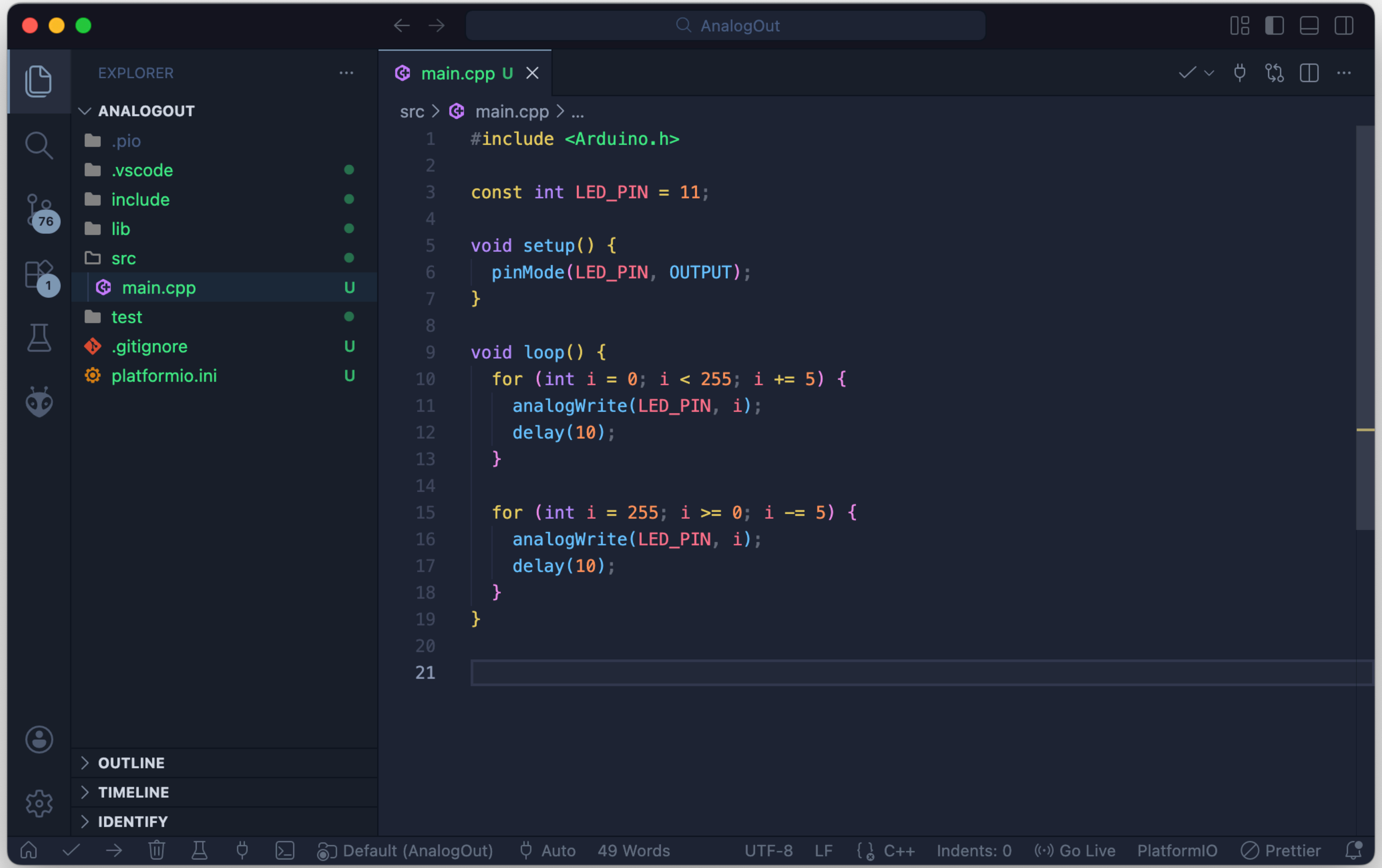
Most Arduino boards can't create true analog signals. We can however fake it, using a technique called PWM via the `analogWrite()` function.



Fade a LED using a loop

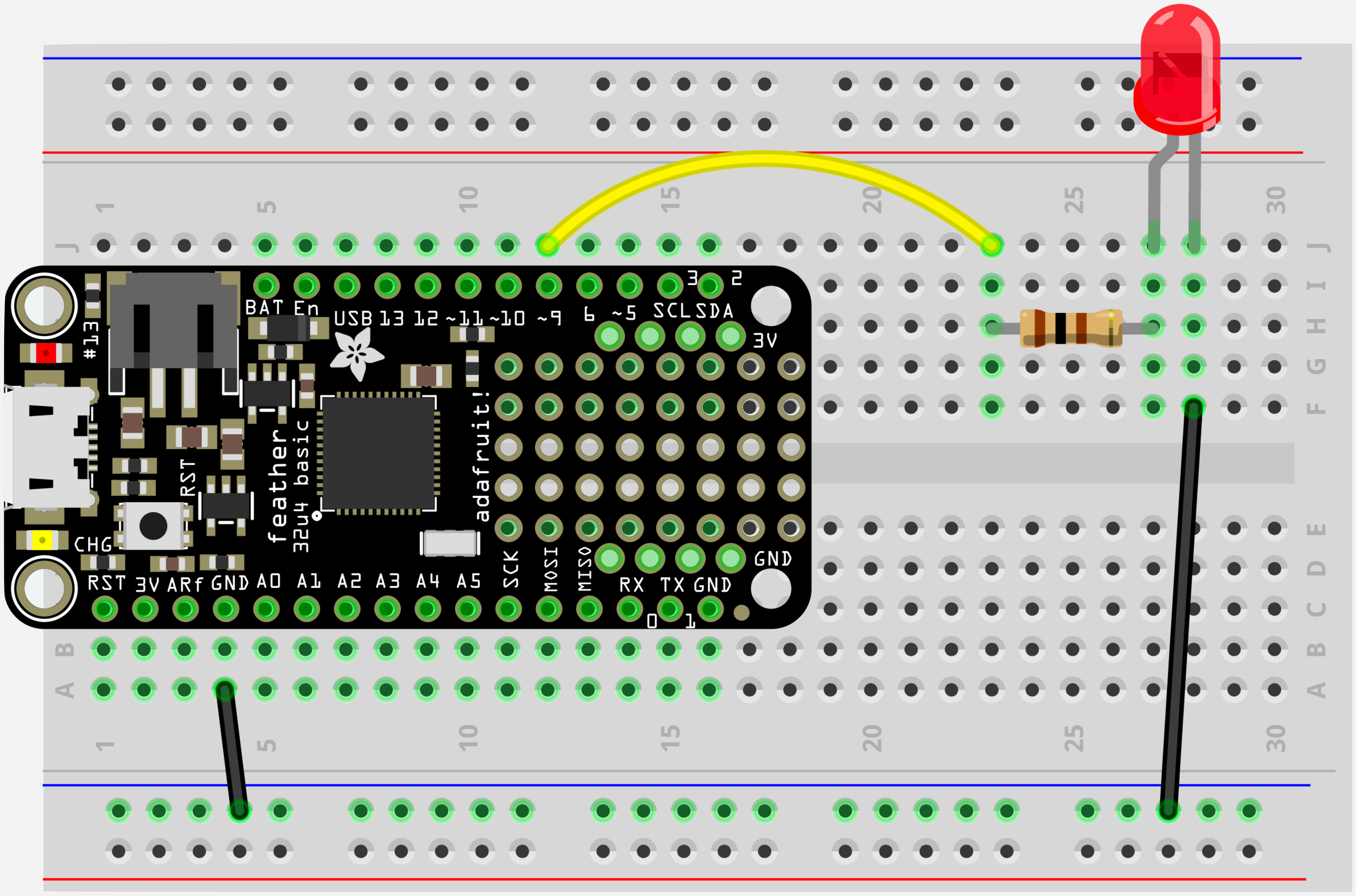


* Resistor is 100Ω (Brown Black Brown)

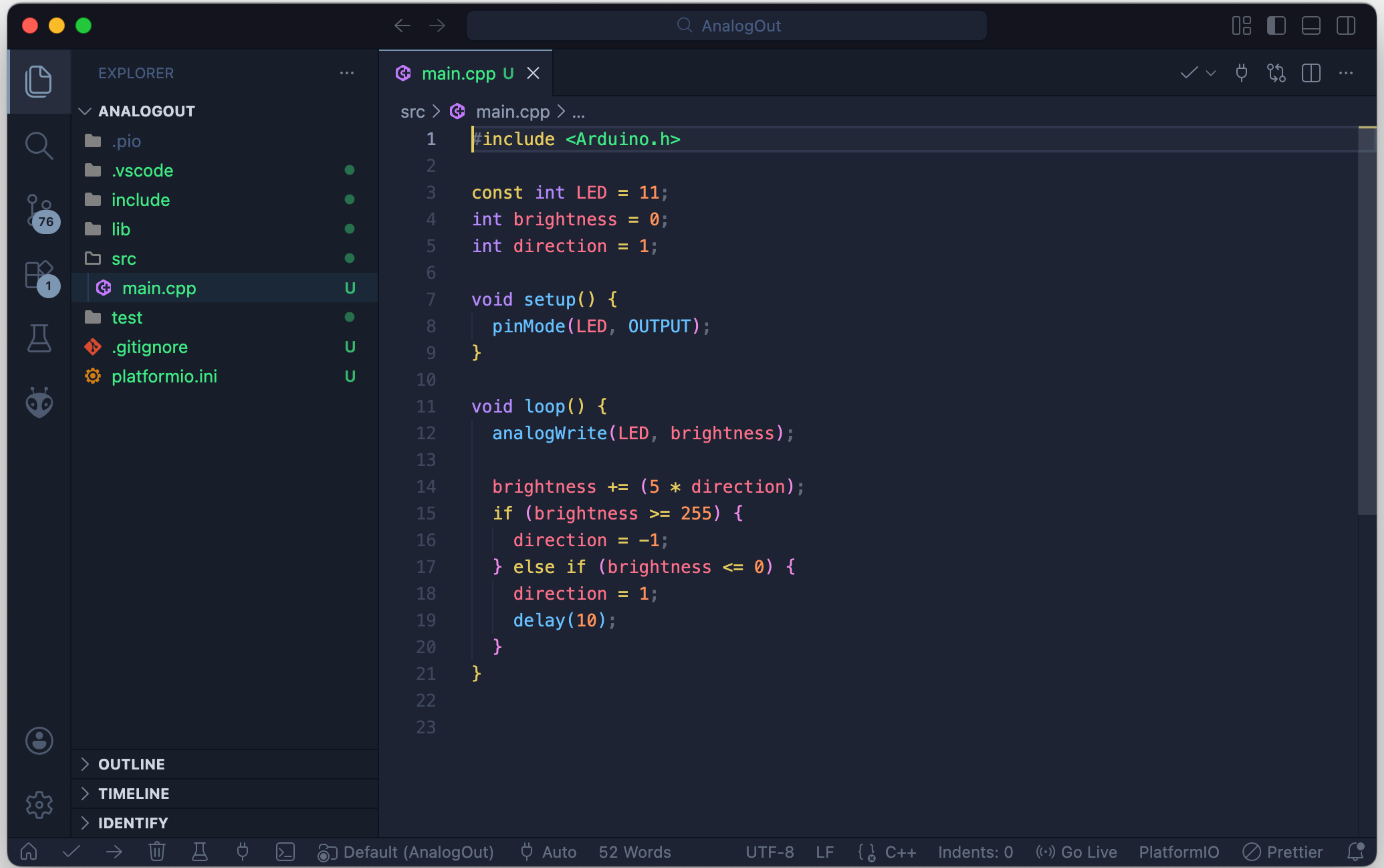


AnalogOut
EXPLORER
ANALOGOUT
.pio
.vscode
include
lib
src
main.cpp
test
.gitignore
platformio.ini
OUTLINE
TIMELINE
IDENTIFY
main.cpp U
src > main.cpp > ...
#include <Arduino.h>
const int LED_PIN = 11;
void setup() {
pinMode(LED_PIN, OUTPUT);
}
void loop() {
for (int i = 0; i < 255; i += 5) {
analogWrite(LED_PIN, i);
delay(10);
}
for (int i = 255; i >= 0; i -= 5) {
analogWrite(LED_PIN, i);
delay(10);
}
}
Default (AnalogOut)
Auto
49 Words
UTF-8
LF
C++
Indents: 0
Go Live
PlatformIO
Prettier

Fade a LED
using a global variable



* Resistor is 100Ω (Brown Black Brown)


AnalogOut
EXPLORER
ANALOGOUT
.pio
.vscode
include
lib
src
main.cpp
test
.gitignore
platformio.ini
OUTLINE
TIMELINE
IDENTIFY
main.cpp U
src > main.cpp > ...
#include <Arduino.h>
const int LED = 11;
int brightness = 0;
int direction = 1;
void setup() {
pinMode(LED, OUTPUT);
}
void loop() {
analogWrite(LED, brightness);
brightness += (5 * direction);
if (brightness >= 255) {
direction = -1;
} else if (brightness <= 0) {
direction = 1;
delay(10);
}
}
Default (AnalogOut)
Auto
52 Words
UTF-8
LF
C++
Indents: 0
Go Live
PlatformIO
Prettier

BASIC AUDIO

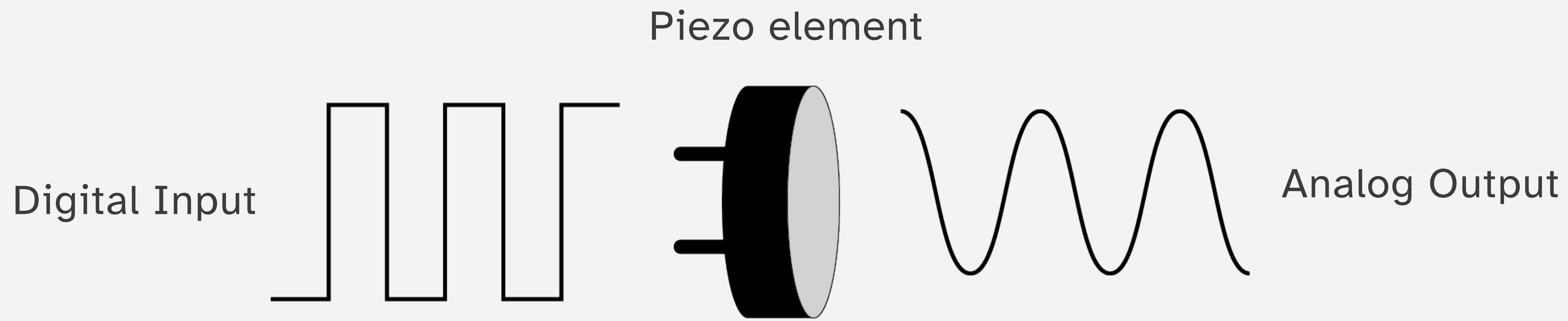
The Piezoelectric Effect

The piezoelectric effect is the ability of certain solid materials to generate an electrical charge when mechanical stress is applied.

This phenomenon involves the conversion of mechanical energy, such as pressure or vibration, into electrical energy.

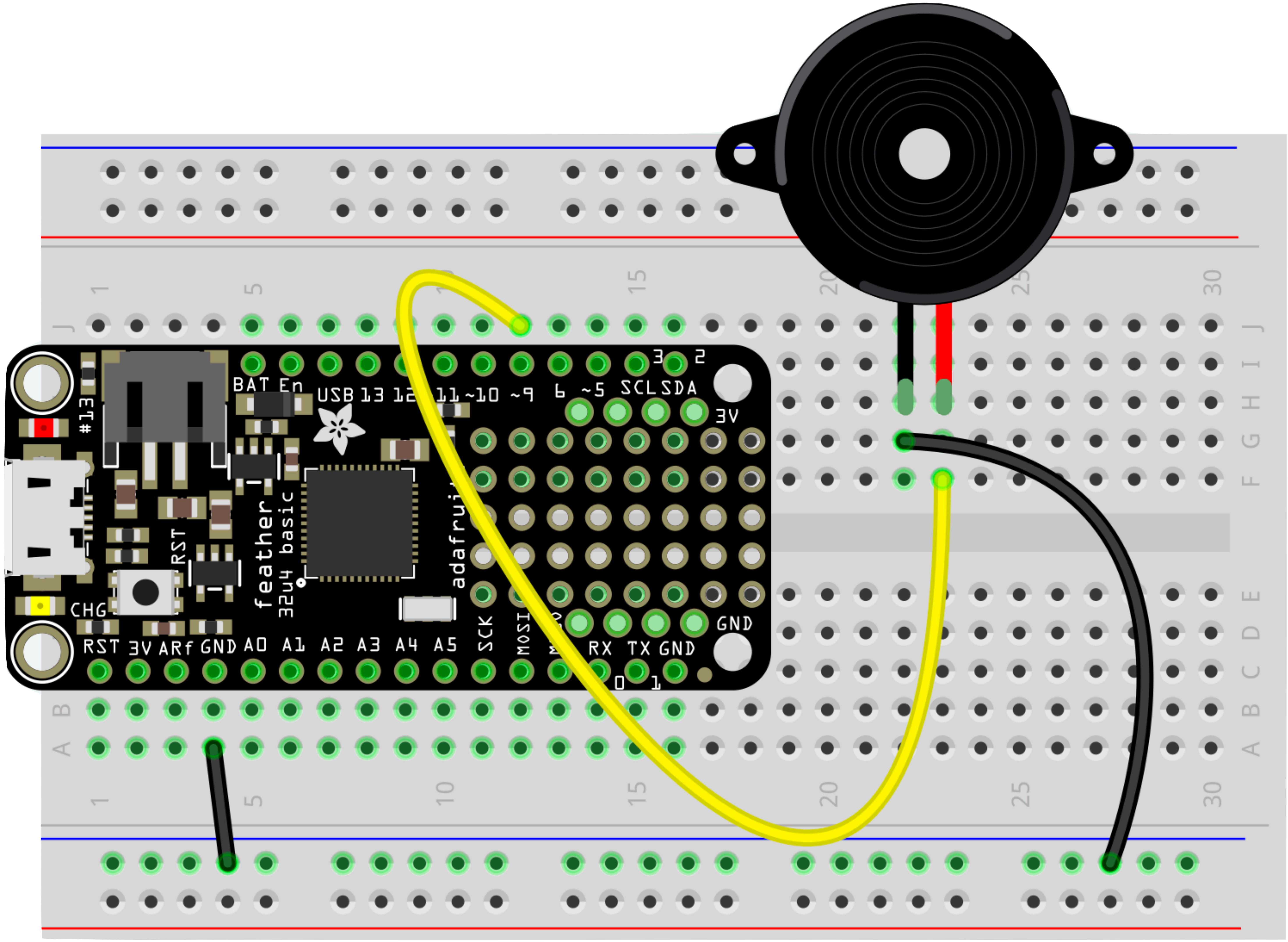
The effect also works in reverse, causing mechanical deformation when an electrical field is applied.

The Inverse Piezoelectric Effect

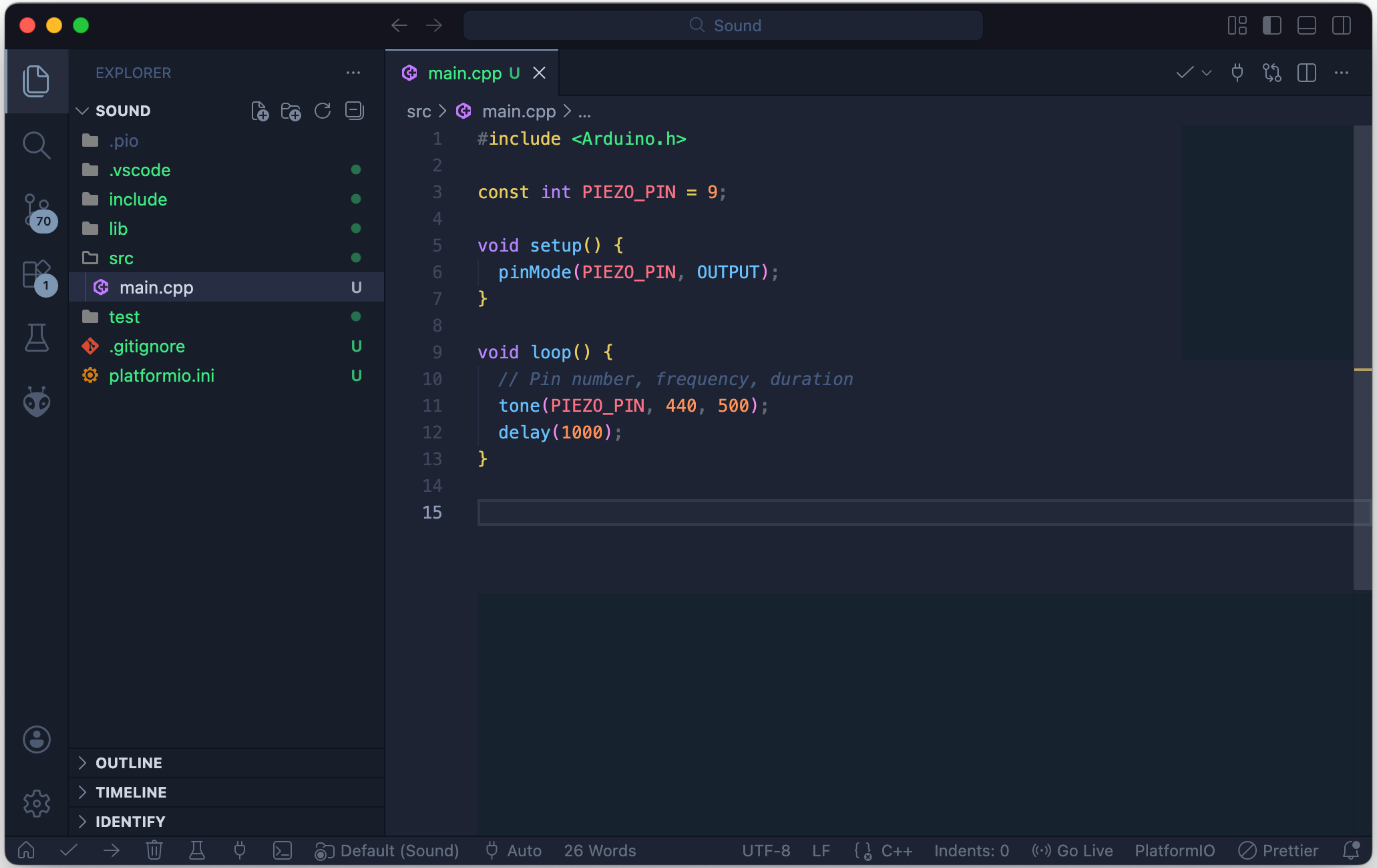


The piezoelectric effect also works in reverse, causing mechanical deformation when an electrical field is applied.

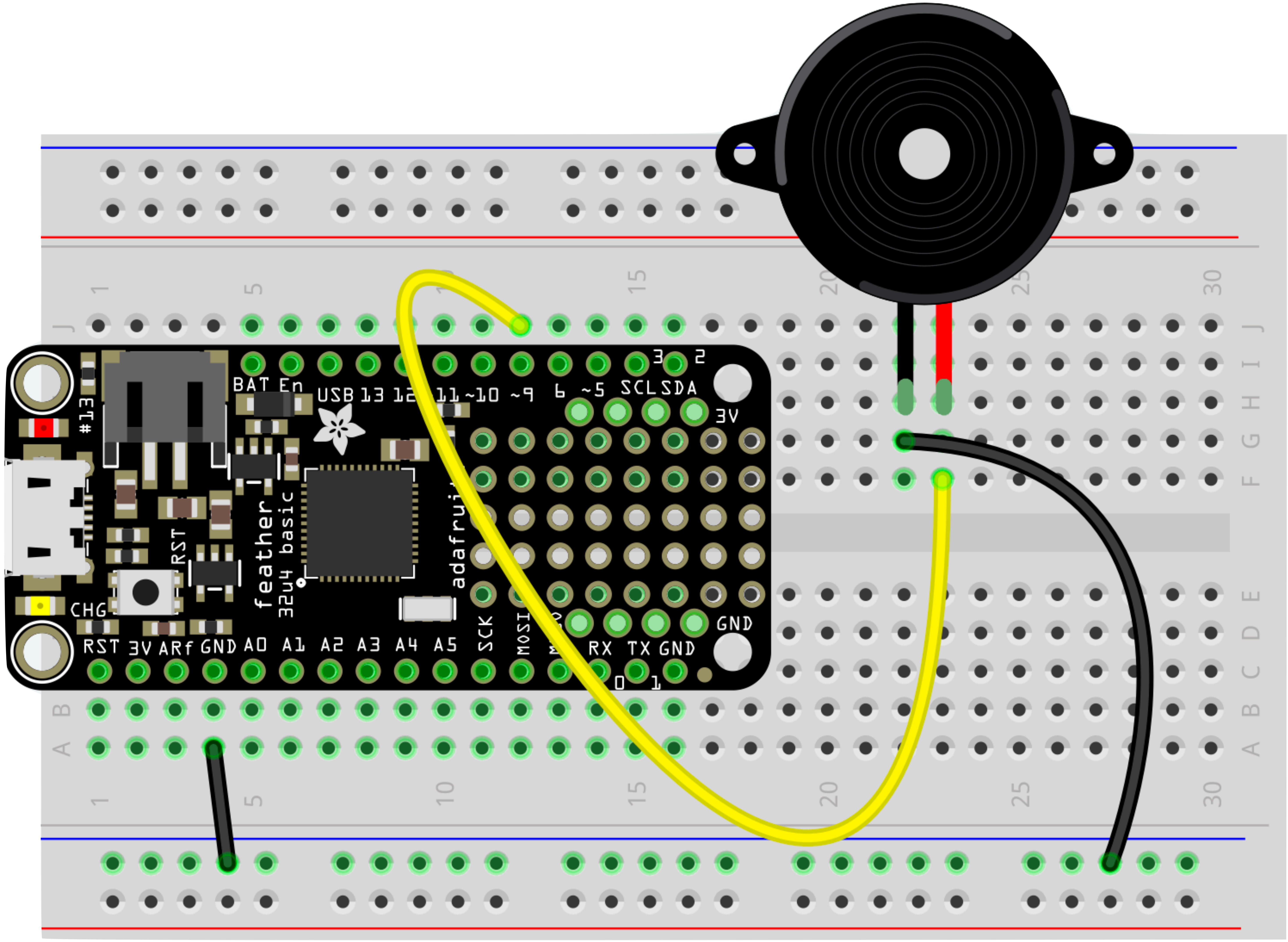
Play a single tone



feather
32u4 basic
adafruit
BAT En USB 13 12 11 ~10 ~9 6 ~5 SCL SDA
3 2
3V
#13
CHG
RST
RST 3V ARf GND A0 A1 A2 A3 A4 A5 SCK MOSI RX TX GND
GND
0 1
1 5 10 15 20 25 30
J I H G F E D C B A

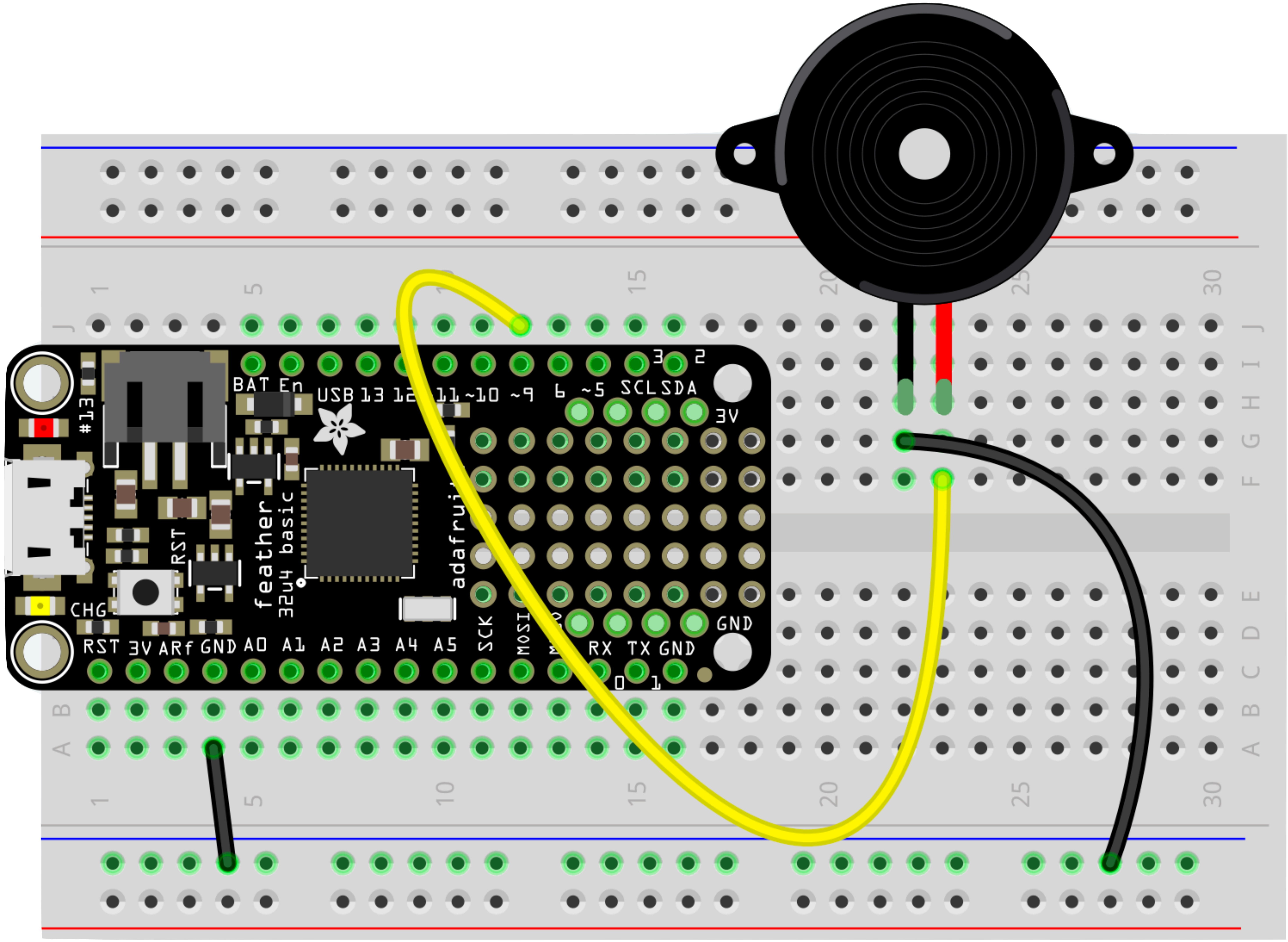

Sound
EXPLORER
SOUND
.pio
.vscode
include
lib
src
main.cpp U
test
.gitignore U
platformio.ini U
OUTLINE
TIMELINE
IDENTIFY
main.cpp U
src > main.cpp > ...
#include <Arduino.h>
const int PIEZO_PIN = 9;
void setup() {
pinMode(PIEZO_PIN, OUTPUT);
}
void loop() {
// Pin number, frequency, duration
tone(PIEZO_PIN, 440, 500);
delay(1000);
}
Default (Sound)
Auto
26 Words
UTF-8
LF
C++
Indents: 0
Go Live
PlatformIO
Prettier

Play a song

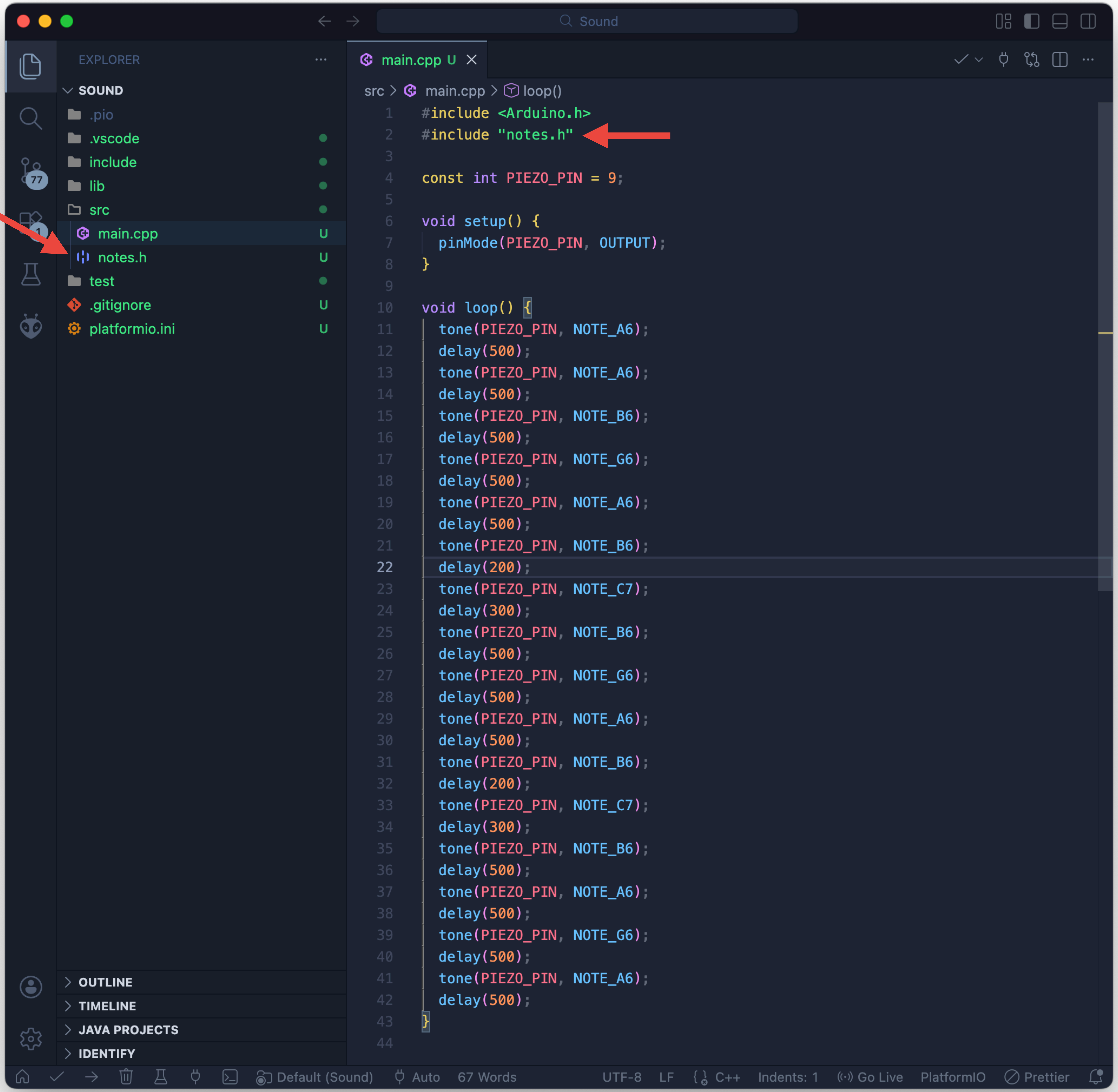

feather
32u4 basic
adafruit
BAT En USB 13 12 11 ~10 ~9 6 ~5 SCL SDA
3 2
3V
GND
RST 3V ARf GND A0 A1 A2 A3 A4 A5 SCK MOSI RX TX GND
0 1
#13
RST
CHG
1 5 10 15 20 25 30
A B C D E F G H I J

```cpp
#include <Arduino.h>

const int PIEZO_PIN = 9;

void setup() {
  pinMode(PIEZO_PIN, OUTPUT);
}

void loop() {
  tone(PIEZO_PIN, 1760);
  delay(500);
  tone(PIEZO_PIN, 1760);
  delay(500);
  tone(PIEZO_PIN, 1976);
  delay(500);
  tone(PIEZO_PIN, 1568);
  delay(500);
  tone(PIEZO_PIN, 1760);
  delay(500);
  tone(PIEZO_PIN, 1976);
  delay(200);
  tone(PIEZO_PIN, 2093);
  delay(300);
  tone(PIEZO_PIN, 1976);
  delay(500);
  tone(PIEZO_PIN, 1568);
  delay(500);
  tone(PIEZO_PIN, 1760);
  delay(500);
  tone(PIEZO_PIN, 1976);
  delay(200);
  tone(PIEZO_PIN, 2093);
  delay(300);
  tone(PIEZO_PIN, 1976);
  delay(500);
  tone(PIEZO_PIN, 1760);
  delay(500);
  tone(PIEZO_PIN, 1568);
  delay(500);
  tone(PIEZO_PIN, 1760);
  delay(500);
}
```

Play a song
using notes.h



feather
32u4 basic
adafruit
BAT En USB 13 12 11 ~10 ~9 6 ~5 SCL SDA
3V
GND
RST 3V ARf GND A0 A1 A2 A3 A4 A5 SCK MOSI RX TX GND
CHG
RST
#13


Sound
EXPLORER
SOUND
.pio
.vscode
include
lib
src
main.cpp U
notes.h U
test
.gitignore U
platformio.ini U
main.cpp U
src > main.cpp > loop()
#include <Arduino.h>
#include "notes.h"

const int PIEZO_PIN = 9;

void setup() {
 pinMode(PIEZO_PIN, OUTPUT);
}

void loop() {
 tone(PIEZO_PIN, NOTE_A6);
 delay(500);
 tone(PIEZO_PIN, NOTE_A6);
 delay(500);
 tone(PIEZO_PIN, NOTE_B6);
 delay(500);
 tone(PIEZO_PIN, NOTE_G6);
 delay(500);
 tone(PIEZO_PIN, NOTE_A6);
 delay(500);
 tone(PIEZO_PIN, NOTE_B6);
 delay(200);
 tone(PIEZO_PIN, NOTE_C7);
 delay(300);
 tone(PIEZO_PIN, NOTE_B6);
 delay(500);
 tone(PIEZO_PIN, NOTE_G6);
 delay(500);
 tone(PIEZO_PIN, NOTE_A6);
 delay(500);
 tone(PIEZO_PIN, NOTE_B6);
 delay(200);
 tone(PIEZO_PIN, NOTE_C7);
 delay(300);
 tone(PIEZO_PIN, NOTE_B6);
 delay(500);
 tone(PIEZO_PIN, NOTE_A6);
 delay(500);
 tone(PIEZO_PIN, NOTE_G6);
 delay(500);
 tone(PIEZO_PIN, NOTE_A6);
 delay(500);
}
OUTLINE
TIMELINE
JAVA PROJECTS
IDENTIFY
Default (Sound)
Auto
67 Words
UTF-8
LF
C++
Indents: 1
Go Live
PlatformIO
Prettier